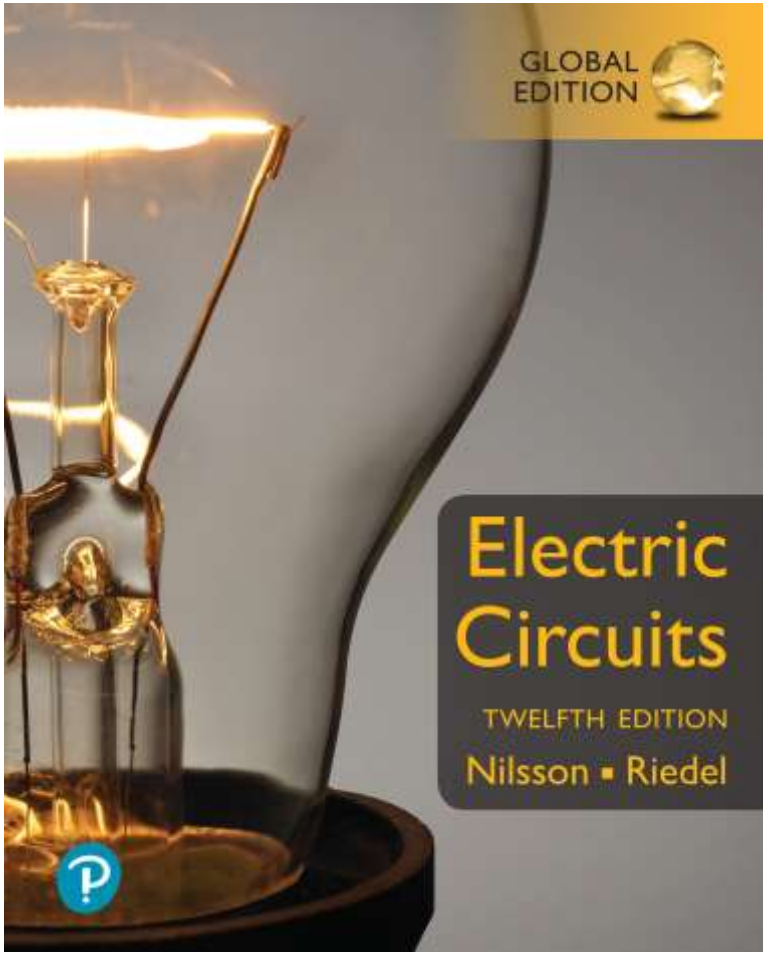
Electric Circuits

Twelfth Edition, Global Edition



Chapter 4 – Part 6

Techniques of Circuit Analysis

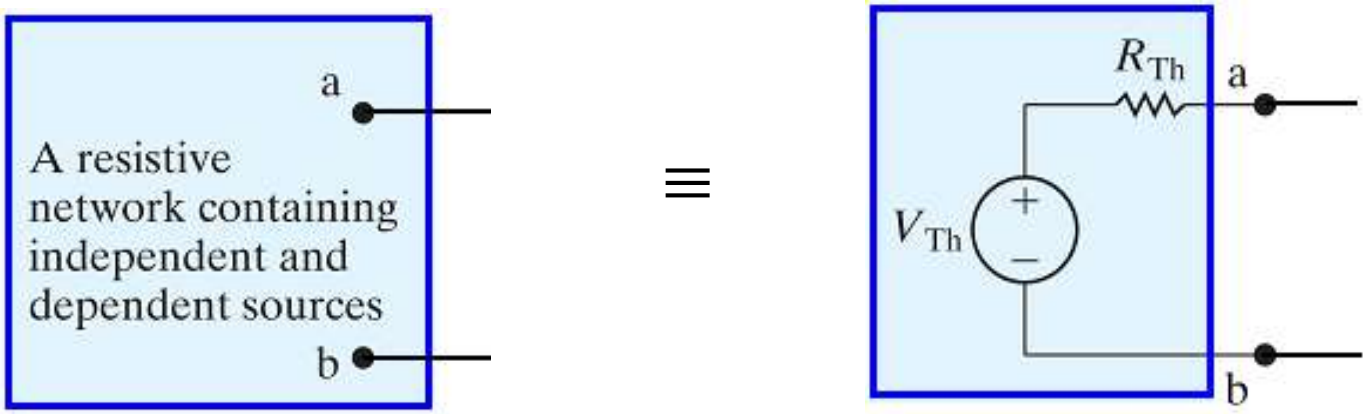
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Chapter 4 – techniques of circuit analysis (1 of 12)

Thevenin and Norton equivalents

- A circuit simplification technique
- Can be applied to any circuit consisting of resistors and sources

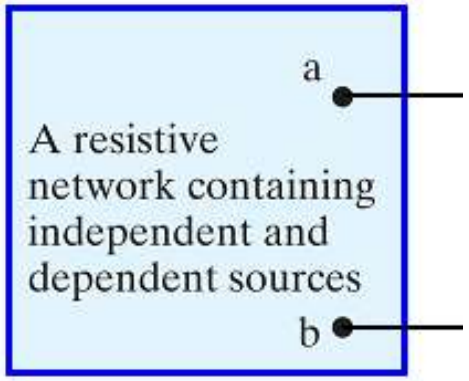


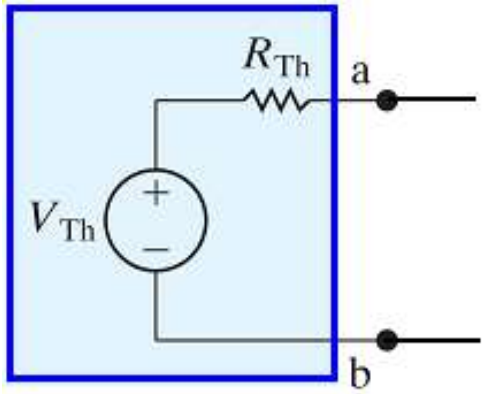
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Chapter 4 – techniques of circuit analysis (2 of 12)

What instrument could we use to determine the voltage across the open circuit in these two circuits?





A. An ohmmeter

B. A voltmeter

C. An ammeter

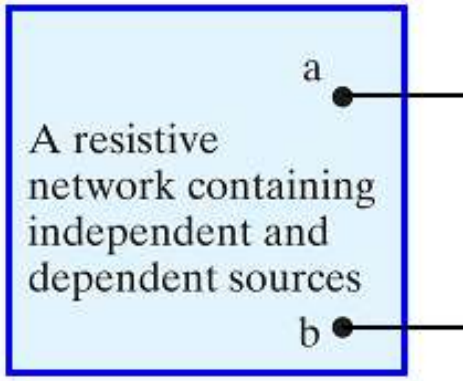
D. None of the above

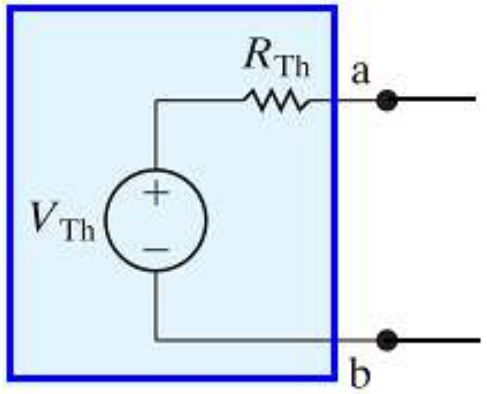
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Chapter 4 – techniques of circuit analysis (2 of 12)

What instrument could we use to determine the voltage across the open circuit in these two circuits?





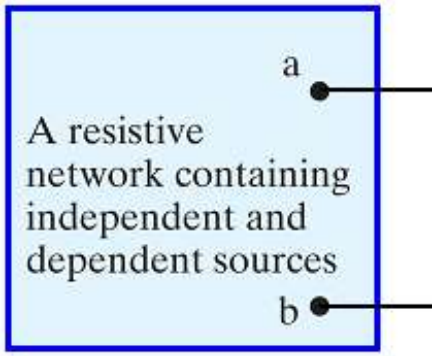
- ✗ A. An ohmmeter
- ✓ B. A voltmeter
- ✗ C. An ammeter
- ✗ D. None of the above

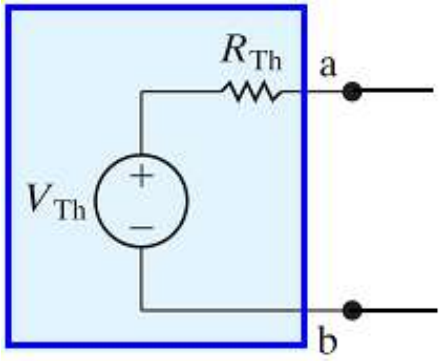
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Chapter 4 – techniques of circuit analysis (3 of 12)

What instrument could we use to determine the current through a short circuit between nodes a and b in these two circuits?





A. An ohmmeter

B. A voltmeter

C. An ammeter

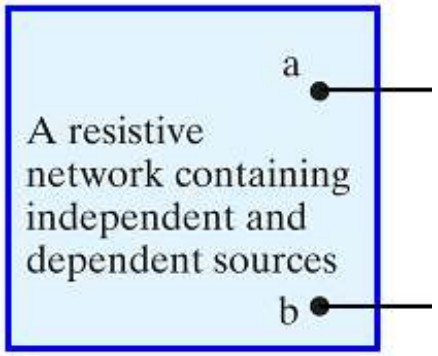
D. None of the above

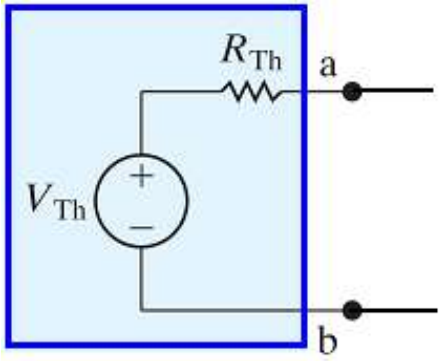
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Chapter 4 – techniques of circuit analysis (3 of 12)

What instrument could we use to determine the current through a short circuit between nodes a and b in these two circuits?





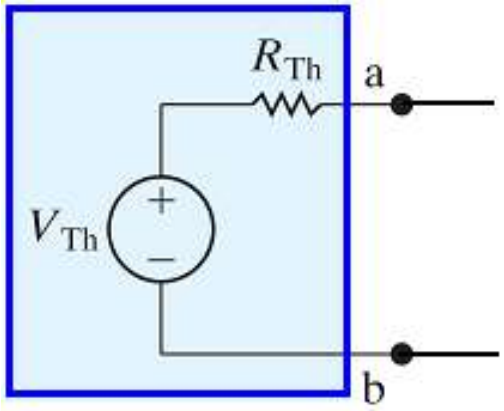
- ✗ A. An ohmmeter
- ✗ B. A voltmeter
- ✓ C. An ammeter
- ✗ D. None of the above

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Chapter 4 – techniques of circuit analysis (4 of 12)

If we create a short circuit at the output, between nodes a and b, what is the current in this short circuit?



A. V_{Th} / R_{Th}

B. $V_{Th}R_{Th}$

C. R_{Th} / V_{Th}

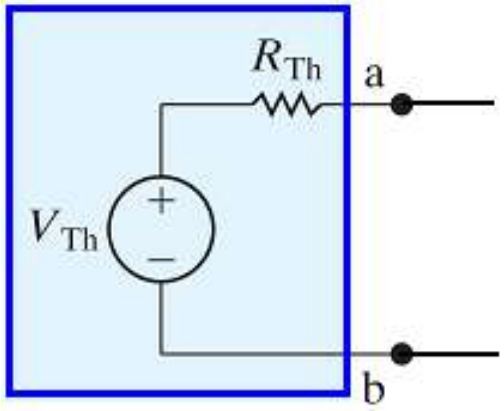
D. None of the above

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Chapter 4 – techniques of circuit analysis (4 of 12)

If we create a short circuit at the output, between nodes a and b, what is the current in this short circuit?



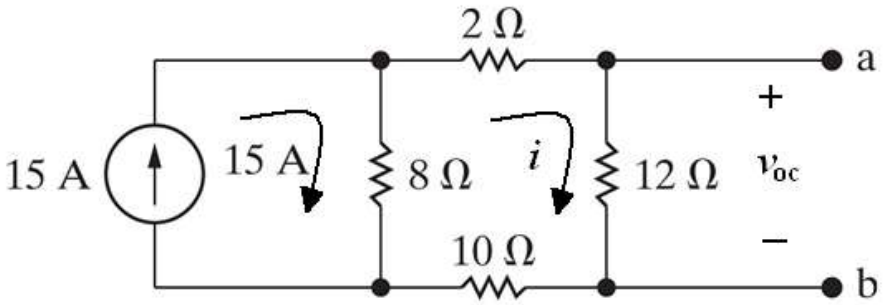
- ✔ A. V_{Th} / R_{Th}
- ✘ B. $V_{Th}R_{Th}$
- ✘ C. R_{Th} / V_{Th}
- ✘ D. None of the above

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Chapter 4 – techniques of circuit analysis (5 of 12)

- Thevenin and Norton equivalents
  - Find the Thevenin equivalent of the circuit below to the left of the terminals a,b.
    - Find the open-circuit voltage:



$$2i + 12i + 10i + 8(i - 15) = 0$$

$$\therefore \qquad i(2 + 12 + 10 + 8) = (8)(15) \qquad \Rightarrow \qquad i = 3.75 \text{ A}$$

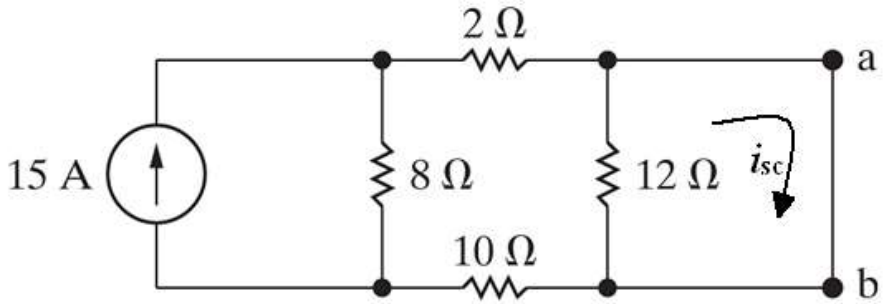
$$v_{oc} = 12i = 45 \text{ V} = V_{Th}$$

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Chapter 4 – techniques of circuit analysis (6 of 12)

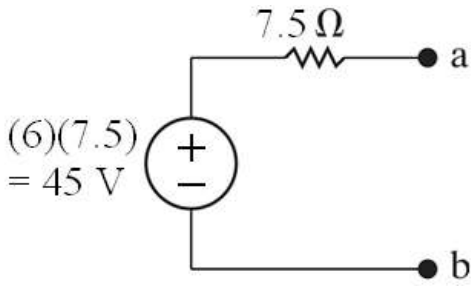
Find the short-circuit current. Here we use current division, noting that the short circuit effectively removes the $12\ \Omega$ resistor from the circuit:



$$i_{sc} = \frac{8 \,||\, (2+10)}{2+10}(15) = 6 \text{ A}$$

$$\therefore \quad R_{Th} = \frac{v_{oc}}{i_{sc}} = \frac{45}{6} = 7.5\Omega$$

The Thevenin equivalent for this original circuit is shown below:

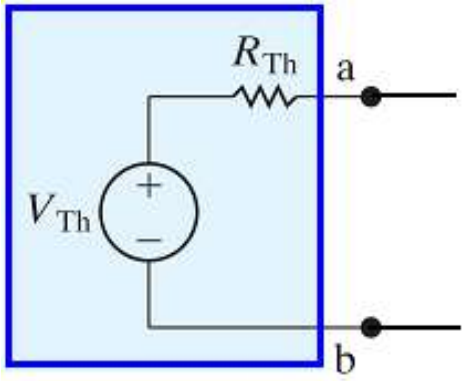


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Chapter 4 – techniques of circuit analysis (7 of 12)

We could use an ohmmeter to measure the Thevenin resistance, so long as there are no dependent sources in the circuit and we remember to



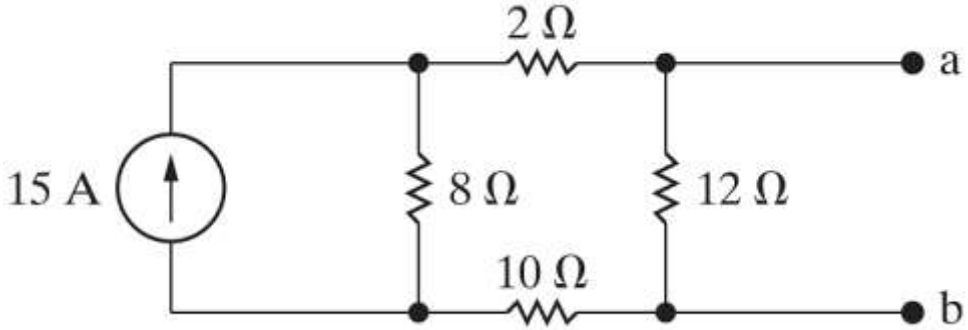
- ✗ A. Create a short at the output
- ✗ B. Place a load resistor at the output
- ✓ C. “Kill” the source by setting its value to zero
- ✗ D. All of the above

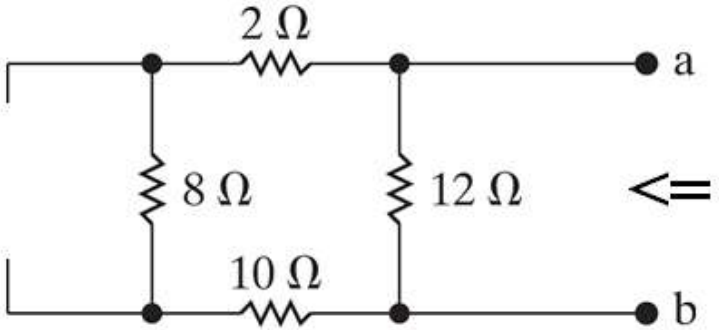
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Chapter 4 – techniques of circuit analysis (8 of 12)

- Thevenin and Norton equivalents
  - Find the Thevenin resistance be killing the source and making series and parallel combinations of resistors.





$$R_{Th} = 12 \,||\, (2+8+10)$$

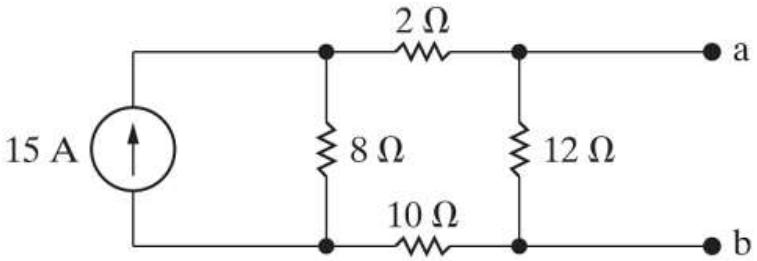
$$= 12 \,||\, 20 = 7.5\Omega$$

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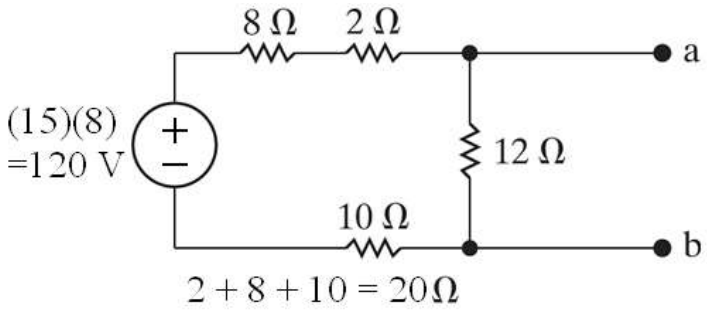
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Chapter 4 – techniques of circuit analysis (9 of 12)

- Thevenin and Norton equivalents
  - An alternative approach – find the Thevenin equivalent of the circuit to the left of the terminals a,b using a series of source transformations.



First step:

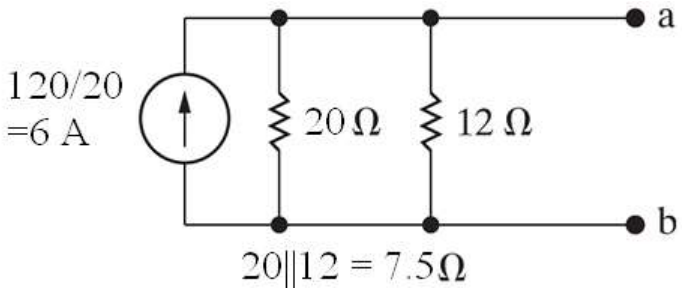


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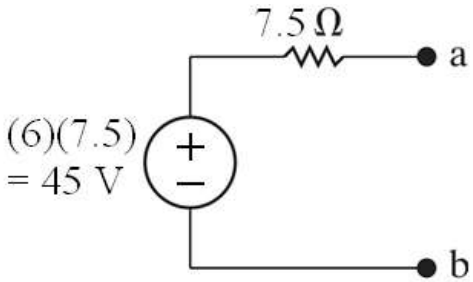
Chapter 4 – techniques of circuit analysis (10 of 12)

- Thevenin and Norton equivalents
  - An alternative approach – find the Thevenin equivalent of the circuit to the left of the terminals a,b using a series of source transformations (continued).

Second step:



Third step:

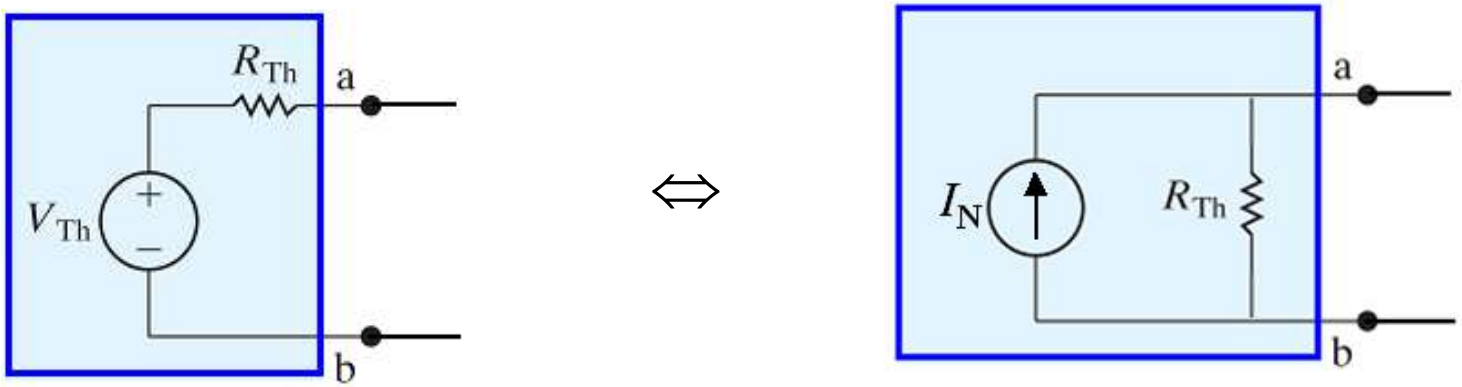


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Chapter 4 – techniques of circuit analysis (11 of 12)

The Norton equivalent is just the source-transform of the Thevenin equivalent. Therefore,



A. $I_N = V_{Th} / R_{Th}$

B. $V_{Th} = R_{Th} I_N$

C. Neither A nor B

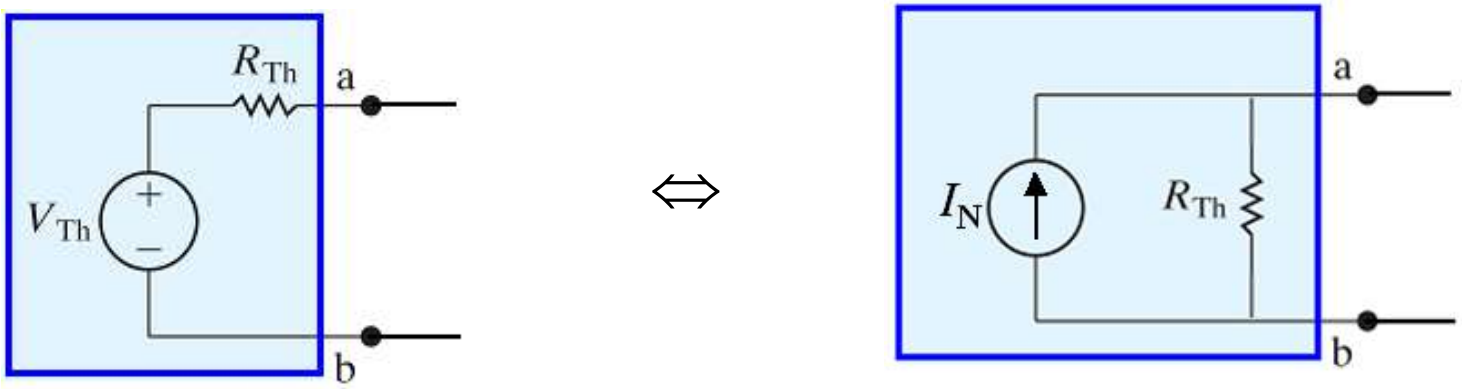
D. Both A and B

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Chapter 4 – techniques of circuit analysis (11 of 12)

The Norton equivalent is just the source-transform of the Thevenin equivalent. Therefore,



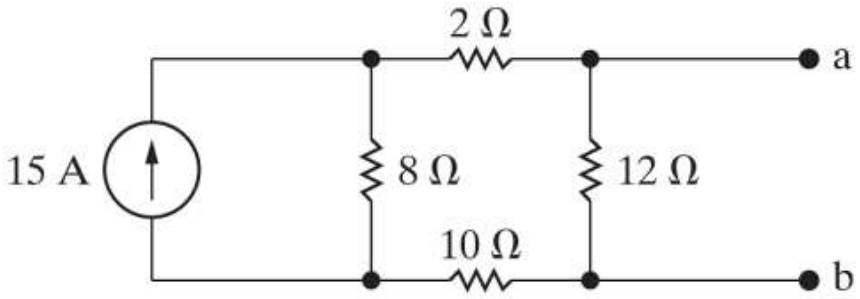
- ✗ A. $I_N = V_{Th} / R_{Th}$
- ✗ B. $V_{Th} = R_{Th} I_N$
- ✗ C. Neither A nor B
- ✓ D. Both A and B

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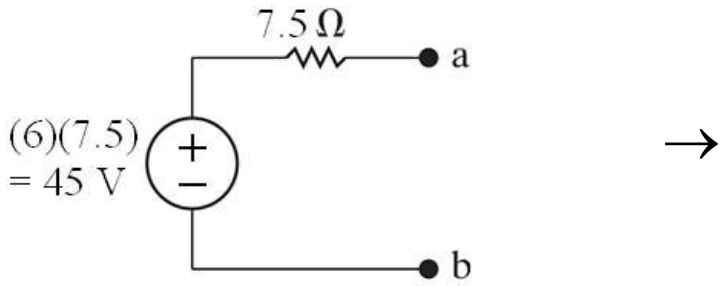
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Chapter 4 – techniques of circuit analysis (12 of 12)

- Thevenin and Norton equivalents
  - Find the Norton equivalent of the circuit to the left of the terminals a,b.



Just source-transform the Thevenin equivalent!



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